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Protective effects of methanol extract of Acori graminei rhizoma and Uncariae Ramulus et Uncus on ischemia-induced neuronal death and cognitive impairments in the rat

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Abstract

Acori graminei rhizoma (AGR) and Uncariae Ramulus et Uncus (URE) have been widely used as herbal medicine against ischemia. In order to investigate whether AGR and URE influenced cerebral ischemia-induced neuronal and cognitive impairments, we examined the effect of AGR and URE on ischemia-induced cell death in the striatum, cortex and hippocampus, and on the impaired learning and memory in the Morris water maze and radial eight-arm maze in rats. After middle cerebral artery occlusion (MCAO) for 2 h, rats were administered saline, AGR or URE (100 mg/kg, p.o.) daily for three weeks, followed by their training to the tasks. In the water maze test, the animals were trained to find a platform in a fixed position during 6 days and then received a 60-s probe trial in which the platform was removed from the pool on the 7th day. In the radial eight-arm maze, animals were tested six times per week for 1 week. Rats with ischemic insults showed impaired learning and memory on the tasks. Pretreatment with AGR and URE produced a significant improvement in escape latency to find the platform in the Morris water maze and in the number of choice errors in the radial arm maze test. Consistent with behavioral data, pretreatments with AGR and URE significantly reduced ischemia-induced cell death in the hippocampal CA1 area. These results demonstrated that AGR and URE have a protective effect against ischemia-induced neuronal loss and learning and memory damage. Our studies suggest that AGR and URE may be useful in the treatment of vascular dementia.

Keywords: Ischemia; Acori graminei rhizoma (AGR); Uncariae Ramulus et Uncus (URE); Learning and memory; Cell death

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Introduction

Ischemia is known to produce severe histopathological damage and related behavioral deficits including cognitive and motor disorders, some of which continue to progress beyond the time of the initial insult. The middle cerebral artery-perfused brain areas such as the parietal cortex, hippocampus, and striatum are mainly affected after cerebral ischemia (Ota et al., 1997; Rice et al., 1981). In particular, the hippocampal neurons, known to play an important role in learning and memory processes (Miyamoto et al., 1987; Morris et al., 1982; Olton et al., 1978), are vulnerable to neuronal injury produced by ischemia (Block et al., 1996; Ota et al., 1997; Schmidt-Kastner et al., 1989). Lesions of hippocampal cells by ischemia are well established to produce severe deficits of learning and memory in a variety of behavioral tasks (Olsen et al., 1994; Sopala and Danysz, 2001).

Acori graminei Rhizoma (AGR) and Uncariae Ramulus et Uncus (URE) have been used clinically as oriental traditional medicines against stroke, Alzheimer's disease or vascular dementia. In particular, AGR, in combination with other herbal drugs, is one of the major components in oriental medical prescriptions for treatment of stroke (Nishiyama et al., 1994b; Zhang et al., 1994). AGR, the dry rhizoma of Acorus gramineus Soland (Araceae), contains volatile oils, which consist mainly of α-asarone (8.8–13.7%) and β-asarone (63.2–81.2%) (Chang and But, 1986). Several studies have demonstrated a variety of pharmacological actions of AGR on the central nervous system. Recently, AGR was reported to produce sedation, decrease locomotor activity, increase pentobarbital-induced sleeping time, and attenuate apomorphine-induced stereotypic behavior in mice (Vohora et al., 1990). AGR was shown to affect the central dopaminergic and GABA systems (Liao et al., 1998). Recent studies have shown that AGR and its major component, asarone, have a neuroprotective effect against excitotoxic neural death (Cho et al., 2000, 2002). In addition, Hsieh et al. reported cognitive enhancing effects of AGR on scopolamine-induced amnesia in rats (Hsieh et al., 2000).

On the other hand, URE is also one of the herbal plants that has long been prescribed for treating stroke and vascular dementia. URE has been described in a medical classic as having the ability to relieve dizziness, headache and tremors resulting from hypertension (Yano, 1987). Many studies have reported potent antihypertensive or vasodilator actions of URE in animal models of hypertension (Goto et al., 1999, 2000; Shi et al., 2003), and that URE or its components had neuroprotective effects against glutamate-induced cell death (Itoh et al., 1999; Shimada et al., 1999) and Ca2+-blocking inhibitory activity (Shi et al., 2003). A recent study demonstrated that URE protected hippocampal neurons against global forebrain ischemia (Suk et al., 2002), suggesting that it might be effective in the treatment of vascular dementia.

In Oriental medicine, many herbal drugs and prescriptions have been used clinically for the treatment of stroke and Alzheimer's disease. Especially, AGR and URE are listed in the Korean Pharmacopoeia and described as two of the most effective medicines useful as sedative and anticonvulsive agents (Huang, 1993). Since AGR and URE have been widely used as herbal medicine against stroke, Alzheimer's disease or vascular dementia, and these drugs have been demonstrated to possess neuroprotective actions, it is possible that these drugs may have beneficial effects on vascular dementia. In the course of screening tests, therefore, we attempted to clarify the effect of these drugs on ischemia-induced impairment of learning and memory using the Morris water maze and the radial eight-arm maze in rats. The neuroprotective effects of these herbal drugs on ischemia-induced cell damage of the striatum, cortex and hippocampus as determined by cresyl violet staining were also examined.

Materials and methods

Animals

Male, Sprague–Dawley rats, weighing 250 ~ 280 g were obtained from Samtaco (Kyungki-do, Korea). Animals were allowed to acclimatize for at least 7 days prior to experimentation. The animals were housed in individual cages under light-controlled conditions and at a room temperature. Food and water were available ad libitum. All experiments were approved by the Kyung Hee University Institutional Animal Care and Use Committee.

Preparation of methanol extract of AGR and URE

AGR and URE were purchased from an oriental drug store (Jungdo Inc. Seoul, Korea). The voucher specimens (No. KH-AGR01 for AGR and No. HP210002 for URE) are deposited at the herbarium located in the College of Oriental Medicine, Kyung Hee University. One hundred grams of ARG and URE were cut into small pieces and extracted three times in a reflux condenser for 24 h with 85% methanol. The solutions were combined, filtered through Whatman No.1 filter paper, and concentrated using a rotary vacuum evaporator followed by lyophilization. The yields of URE and AGR were 12.1 and 10.1%(w/w), respectively.

Focal ischemia model

Focal cerebral ischemia was induced using the intraluminal filament technique. Anesthesia was induced with 3% isoflurane in 30% O_2 / 70% N_2O and then maintained throughout the operation with 0.5% to 0.6% isoflurane delivered via a nose mask. The right common carotid artery was exposed through a midline cervical incision. A heparinized intraluminal filament (ϕ 0.28 mm, rounded tip) was introduced via the external carotid artery. Rectal temperature was monitored and maintained at 37 °C using a heating pad (Harvard Homeothermic Blanket Control Unit, 50-7061). After 120 min of occlusion, the filament was gently pulled out and the external carotid artery was permanently closed by cauterization. In sham-operated rats, the right common carotid artery was exposed and external carotid artery was opened without introducing the filament into the internal carotid artery. After the operation, the animals were allowed to wake up to gain consciousness in an incubator (30 °C) and were then moved to their home cages.

Experimental design

AGR and URE were dissolved in 0.9% NaCl and administered to the AGR + ISCH and URE + ISCH groups, respectively, at a dose of 100 mg/kg (p.o.) daily for 3 weeks after induction of ischemia. The SAL + ISCH group was injected with 0.9% saline (p.o.) after induction of ischemia. The sham-operated control group (SHAM) was treated only with 0.9% saline (p.o.) for 3 weeks. The water maze and radial arm maze tests were performed 4 weeks after induction of ischemia.

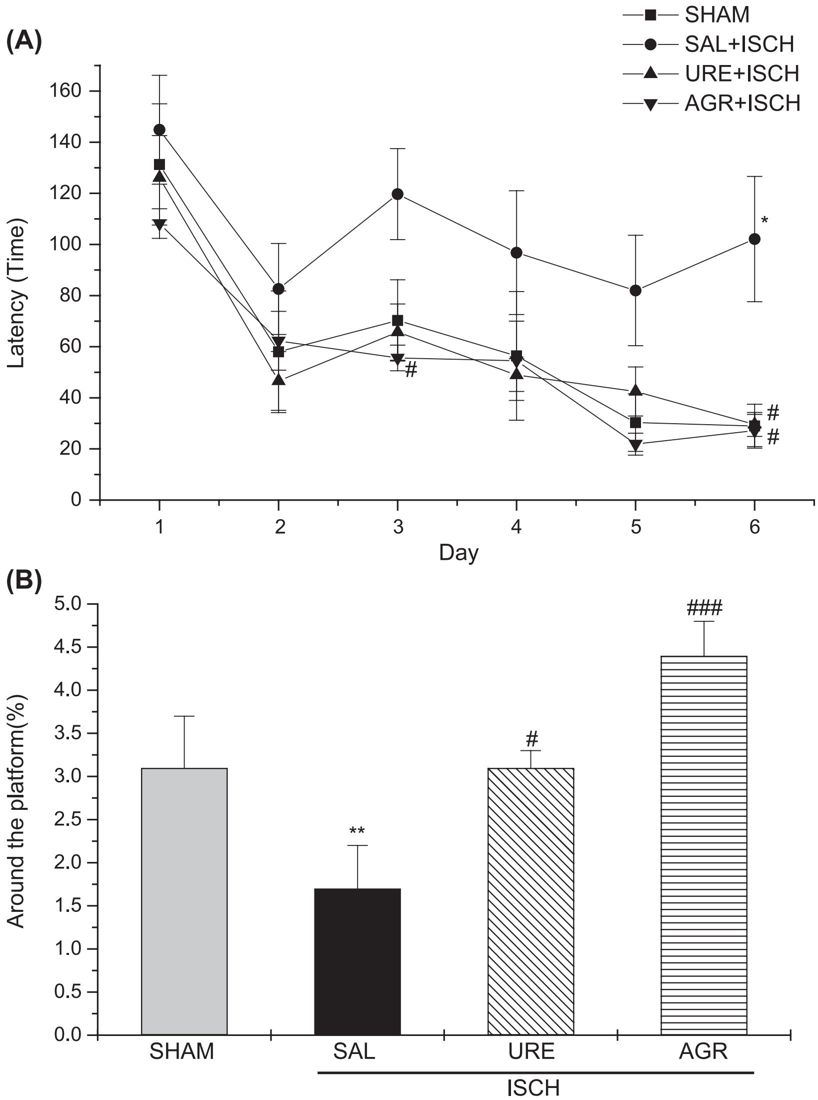

Fig. 1. Effect of AGR and URE on cerebral ischemia-induced deficits in learning and memory in rats performing the Morris water maze. The task was performed with four trials per day during 6 days for the acquisition test (A), and performed with four trials on day 7 without the platform for the retention test (B). Rats were pretreated with saline (1 ml/kg, • SAL + ISCH, n = 7), AGR (100 mg/kg, p.o, ▾ AGR + ISCH, n = 5), URE (100 mg/kg, p.o, ▲ URE + ISCH, n = 5) daily for 3 weeks after cerebral ischemia. The sham-operated control group (■ SHAM, n = 6) was treated with only saline without induction of ischemia. Significance with Tukey's test following a repeated ANOVA is indicated as $^{*}P < 0.05$ and $^{**}P < 0.01$ vs Sham-operated group, or $^{\#}P < 0.05$ and $^{\#\#\#}P < 0.001$ vs SAL + ISCH group. Vertical lines indicate S.E.M (N = 5–7).

Behavioral tests

Exp 1: water maze task

The swimming pool of the Morris water maze was a circular water tank, 200 cm in diameter and 35 cm deep. It was filled to a depth 21 cm with water at 23 ± 2 °C. A platform, 15 cm in diameter and 20

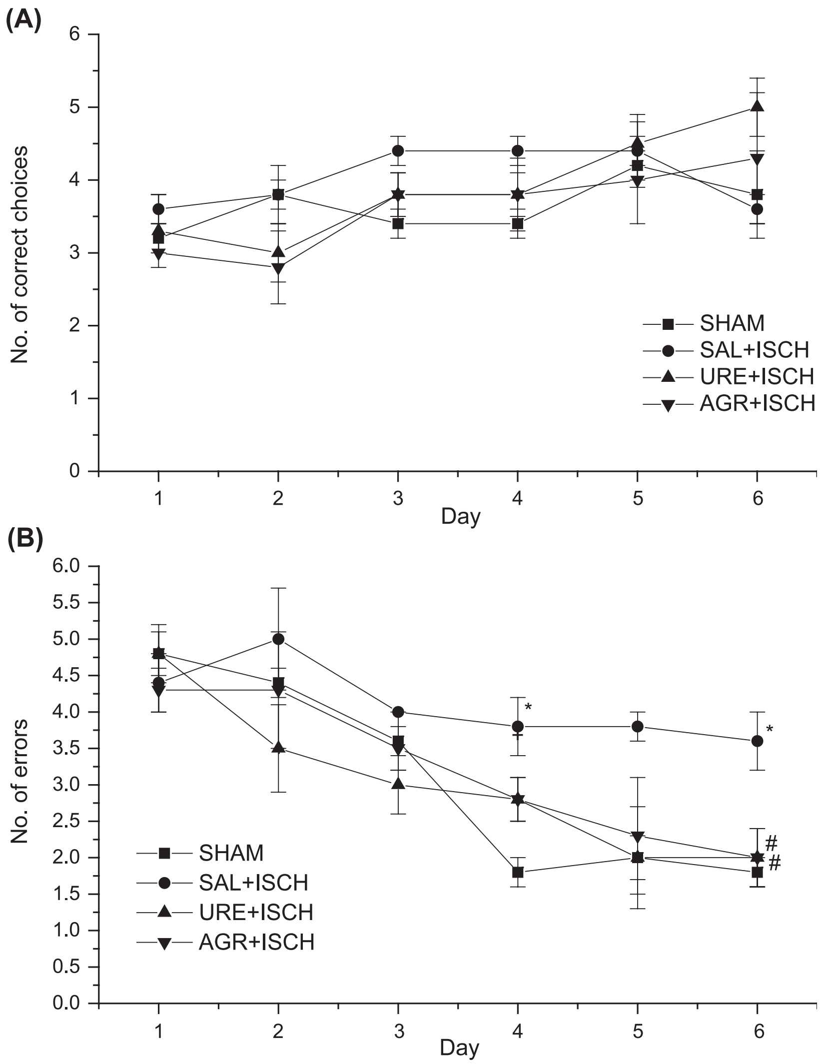

Fig. 2. Effects of AGR and URE on choice accuracy of the eight-arm radial maze task in ischemia-induced rats (A; Number of correct choices, B; Number of errors). The task was started on the 6th week after ischemia and was performed with four trials per day for 6 days. Rats were pretreated with saline (1 ml/kg, • SAL + ISCH, n = 5), AGR (100 mg/kg, p.o, ▼ AGR + ISCH, n = 4), URE (100 mg/kg, p.o, ▲ URE + ISCH, n = 4) for 3 weeks after cerebral ischemia. The sham-operated control group (■ SHAM, n = 5) was treated with only saline without induction of ischemia. Significance with Tukey's test following a repeated ANOVA is indicated as $^{*}P < 0.05$ vs Sham-operated group, or $^{\#}P < 0.05$ vs SAL + ISCH group. Vertical lines indicate S.E.M. (N = 4–5).

cm in height, was placed inside the tank, its top surface being 1.5 cm below the surface of the water. The pool was surrounded by many cues external to the maze. A CCD camera was equipped with a personal computer for behavioral analysis. Each rat received four trials daily for 7 consecutive days. In each trial, the latency to escape onto the hidden platform was recorded within 180 s. The animals were trained to find a platform in a fixed position during 6 days of the acquisition test and then received a 60-s probe trial in which the platform was removed from the pool on the 7th day for the retention test. The intertrial interval time was $\sim$ 1 min. Performance of the test animal in each water maze trial was assessed by use of a personal computer for behavioral analysis (S-mart program, Spain).

Exp 2: eight-arm radial maze task

The maze consisted of a central platform 50 cm in diameter, with eight arms extending radially. Each arm was 70 cm in length and 10 cm wide and with wooden sidewalls 25 cm in height. Food cups for reinforcement were located in a room containing many extra-maze visual cues. Task performance of rats in the eight-arm radial maze was quantified by an image motion analysis program (S-mart, Spain). The trial continued until the test animals entered all eight arms or 5 min had elapsed. The task was performed

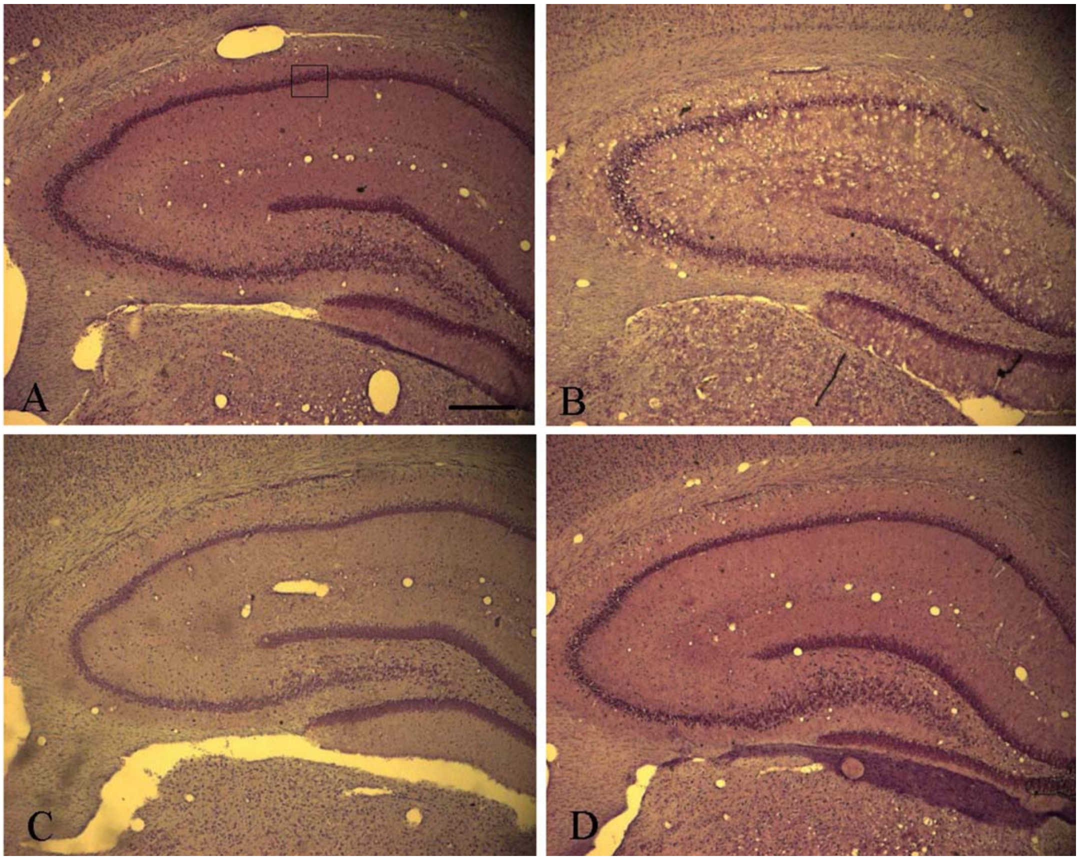

Fig. 3. Representative photographs showing the hippocampus of SHAM (A), SAL + ISCH (B), AGR + ISCH (C) and URE + ISCH groups (D). Low (Fig. 3) and high (Fig. 4) power photomicrographs of the same fields of the hippocampus. Scale bars represent 200 μm (Fig. 3) and 50 μm (Fig. 4), respectively.

with four trials per day for 6 days. Performance of the test in each trial was assessed by three parameters: the number of correct choices in the initial eight chosen arms, the number of errors which was defined as choosing arms which had already been visited, and the running time elapsed before the animal consumed all eight pellets.

Cresyl violet-staining

After behavioral testing was completed, all animals were deeply anesthetized with sodium pentobarbital (80 mg/kg, i.p.) and then perfused through the ascending arota with normal saline (0.9%), followed by 500 ml of 4% paraformaldehype in 0.1 M phosphate buffer. The brains were removed, postfixed overnight and cryoprotected in 20% sucrose in 0.1 M phosphate buffer saline (PBS) at 4 °C. The brains were cut with a cryostat into 30 μm coronal sections, then processed histochemically as free-floating sections. All sections of the brain were mounted on gelatine-coated microscope slides, allowed to dry for 10 min and stained with cresyl violet for 5 min. They were then dehydrated in increasing ethanol concentrations and finally coverslipped from xylene. Hippocampal cells were measured by placing the

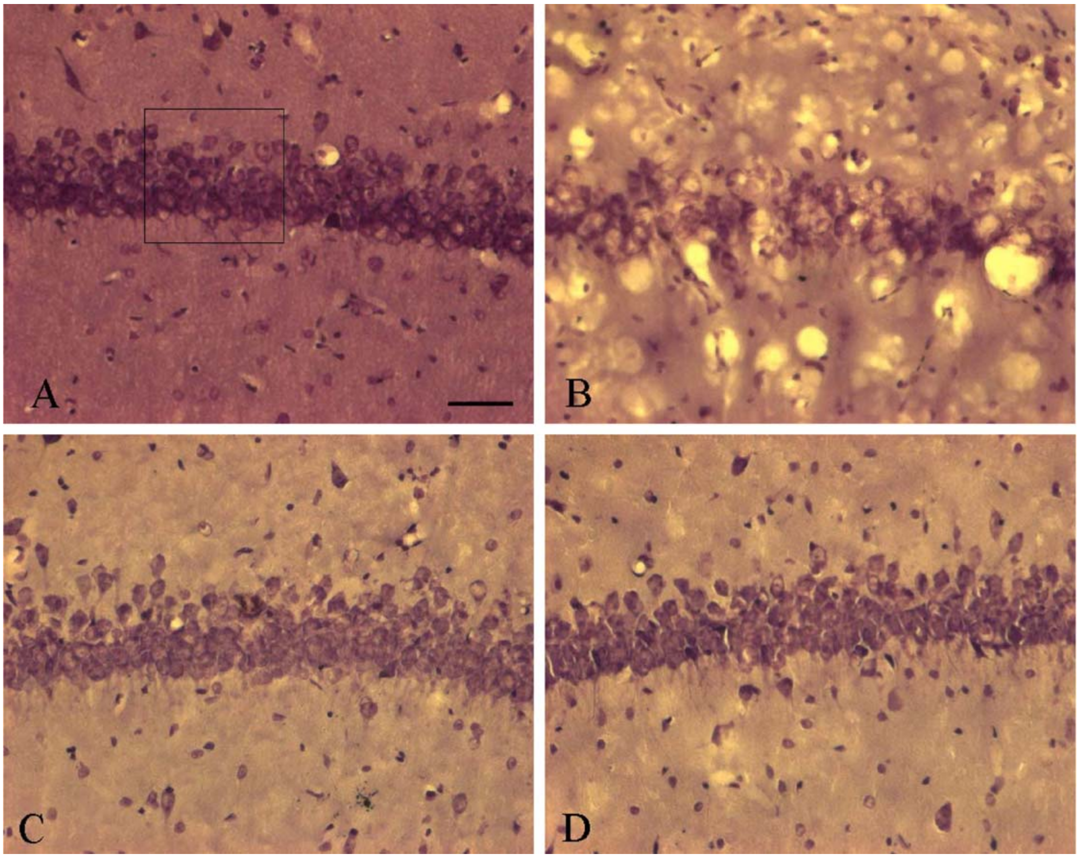

Fig. 4. Representative photographs showing the hippocampus of SHAM (A), SAL + ISCH (B), AGR + ISCH (C) and URE + ISCH groups (D). Low (Fig. 3) and high (Fig. 4) power photomicrographs of the same fields of the hippocampus. Scale bars represent 200 μm (Fig. 3) and 50 μm (Fig. 4), respectively.

grid on the hippocampal CA1 area at the level of 4.8 mm bregma according to the atlas of Paxinos and Watson (Paxinos and Watson, 1986). The cells of the hippocampal CA1 area were counted at 400 × magnification using a microscope rectangle grid measuring 100 × 100 microns. The cells of the dorsolateral striatum and the parietal cortex were counted at 200 × magnification using a microscope rectangle grid measuring 200 × 200 microns. Evaluation of cell loss was determined by calculating the density of cells using the Scion image system (Scion Corp., Maryland, U.S.A.). Cells within the examined area were counted on each of 3 sections per animal.

Statistical analysis

Results were expressed as means ± S.E. The behavioral data were analyzed using ANOVA for repeated measures. Statistical differences among groups were further analyzed using Tukey's post hoc test. Histological data were calculated and analyzed by one-way ANOVA followed by the Tukey's post hoc technique.

Results

Exp 1: water maze test

Forebrain ischemia affected the performance of rats in the water maze (Fig. 1). The SAL + ISCH group showed poorer performance than did the SHAM group. As seen in Fig. 1A, the latencies to find

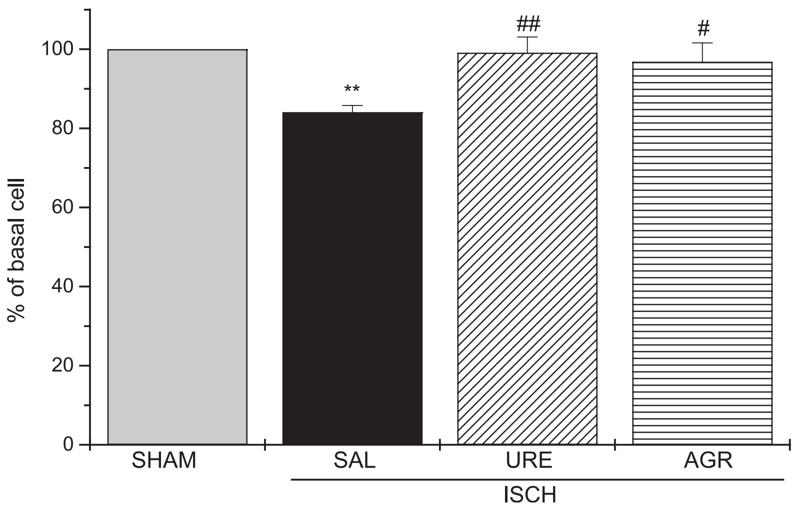

Fig. 5. Effects of AGR and URE on ischemia-induced neuronal cell damage in the hippocampus. The density of hippocampal CA1 cell of cresyl violet-stained sections was measured within a 100 × 100 μm grid over the areas at 400 × magnification. Rats were pretreated with saline (1 ml/kg, SAL + ISCH), AGR (100 mg/kg, p.o., AGR + ISCH), URE (100 mg/kg, p.o., URE + ISCH) daily for 3 weeks after cerebral ischemia. The sham-operated control group (SHAM) was treated with only saline without induction of ischemia. Significance with Tukey's test following a repeated ANOVA is indicated as $^{**}P < 0.01$ vs Sham-operated group, or $^{\#}P < 0.05$ and $^{\#\#}P < 0.01$ vs SAL + ISCH group. Vertical lines indicate S.E.M. (N = 5–7).

the hidden platform by the AGR + ISCH and URE + ISCH groups were significantly decreased compared to those of the SAL + ISCH group. An ANOVA (4 × 6, treatment × time) performed on the swimming time of acquisition trials revealed significant effect of a group difference ($F(3,19) = 3.689$, $P < 0.05$), effect of day ($F(5,95) = 26.259$, $P < 0.001$), but not group × day interaction ($F(15,95) = 1.116$, $P = 0.353$). Tukey's post-hoc test revealed that AGR + ISCH ($P < 0.05$ on day 3 and 6, respectively) and URE + ISCH groups ($P < 0.05$ on day 6) significantly reduced the latency of swimming time, compared with those of the SAL + ISCH group. On the 7th day, post-hoc test on retention performance also revealed that the AGR + ISCH ($P < 0.001$) and URE + ISC ($P < 0.05$) groups spent longer times around the platform compared with the SAL + ISCH group (Fig. 1B). Ischemia severely impaired spatial cognition in the water maze task. AGR and URE groups displayed attenuated ischemia-induced learning and memory damage in the water maze.

Exp 2: eight-arm radial maze test

The SAL + ISCH group showed a slower acquisition curve than the SHAM group as shown in Fig. 2A. The SAL + ISCH group showed a greater number of errors over testing as compared with the SHAM group. AGR + ISCH and URE + ISCH groups showed significantly reduced number of errors

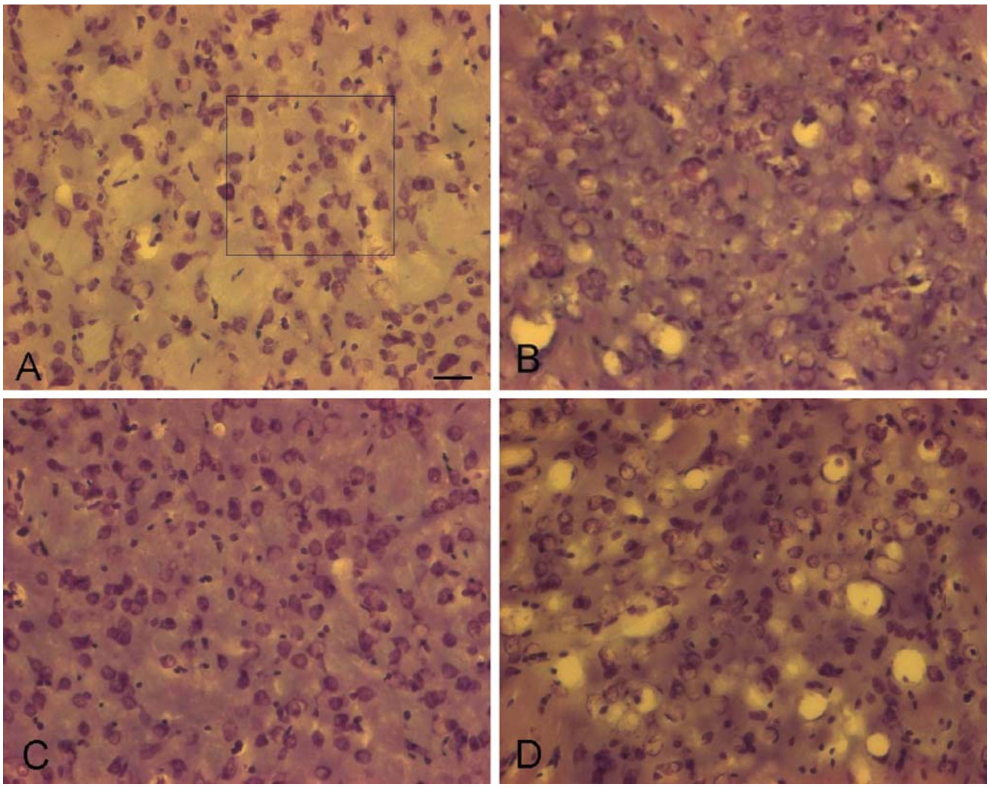

Fig. 6. Representative photographs showing the striatum of SHAM (A), SAL + ISCH (B), AGR + ISCH (C) and URE + ISCH groups (D). Scale bars represent 100 μm.

compared with the SAL + ISCH group as seen in Fig. 2B. An ANOVA (4×6, treatment X time) performed on the number of errors indicated a significant difference among the groups ($F(3, 14) = 14.515$, $P < 0.001$), a significant effect of day ($F(5,70) = 18.827$, $P < 0.001$), but not a group $\times$ day interaction ($F(15, 70) = 1.322$, $P = 0.213$). Post-hoc analysis revealed that administration of AGR and URE produced a progressive decrease in number of error response compared with that of the SAL + ISCH group. These recovering effects of the drugs AGR and URE were significant after the 6th dose ($p < 0.05$ for both groups), as shown in Fig. 2B. Performance in the eight-arm radial maze task was impaired in the SAL + ISCH group, and AGR and URE treatment attenuated ischemia-induced learning and memory damage in the eight-arm radial maze.

Brain histology

The brains of animals that were tested on the behavioral tasks were analyzed histologically for the extent of neuronal cell loss. Brains of the SAL + ISCH group showed a significant cell loss of the hippocampal CA1 area in contrast with the SHAM group. Pretreatment with AGR and URE significantly prevented ischemia-induced cell loss in the hippocampal CA1 area as seen in Figs. 3, 4 and 5. Measures of one-way ANOVA on the density of cells revealed a significant difference among

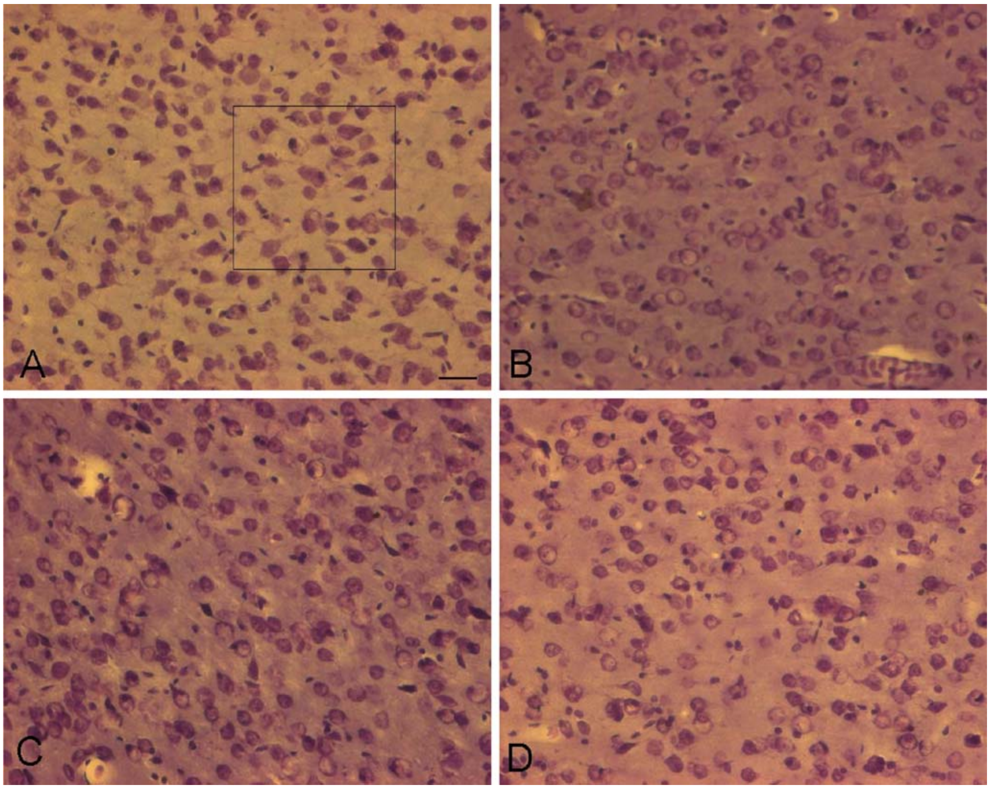

Fig. 7. Representative photographs showing the cortex of SHAM (A), SAL + ISCH (B), AGR + ISCH (C) and URE + ISCH groups (D). Scale bars represent 100 μm.

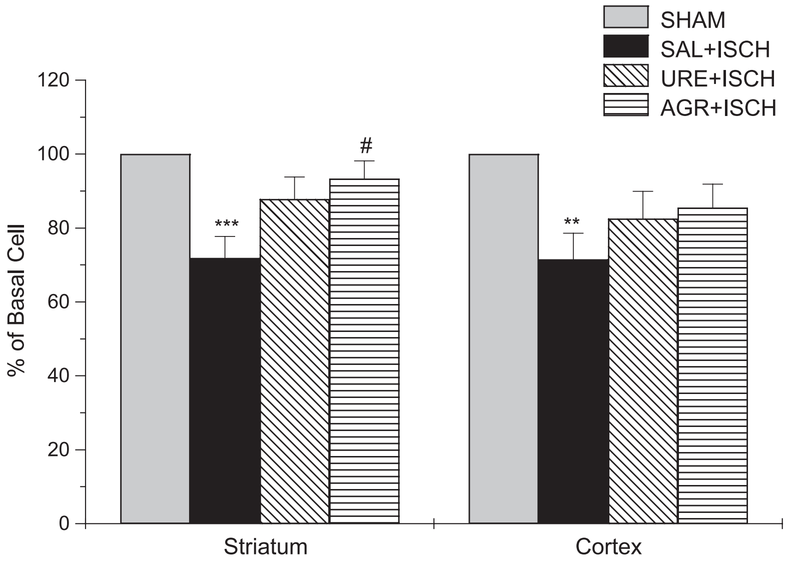

Fig. 8. Effects of AGR and URE on ischemia-induced neuronal cell damage in the striatum and cortex. The density of striatum and cortex cell of cresyl violet-stained sections was measured within a 200 × 200 μm grid over the areas at 200× magnification. Rats were pretreated with saline (1 ml/kg, SAL + ISCH), AGR (100 mg/kg, p.o., AGR + ISCH), URE (100 mg/kg, p.o., URE + ISCH) for 3 weeks after cerebral ischemia. The sham-operated control group (SHAM) was treated with only saline without induction of ischemia. Significance with Tukey's test following a repeated ANOVA is indicated as $^{**}P < 0.01$ and $^{***}P < 0.001$ vs Sham-operated group, or $^{\#}P < 0.05$ vs SAL + ISCH group. Vertical lines indicate S.E.M. (N = 5–7).

the groups ($F(3,68) = 7.327$, $P < 0.001$). Post-hoc comparisons revealed that URE + ISCH ($P < 0.01$) and AGR + ISCH groups ($P < 0.05$) had more cells in the hippocampus compared with those of the SAL + ISCH group.

The SAL + ISCH group showed significant cell loss of the striatum (Fig. 6) and the parietal cortex (Fig. 7) in contrast with the SHAM group. Pretreatment with AGR significantly prevented ischemia-induced cell loss in the striatum, but not in the cortex area as seen in Figs. 6, 7 and 8. Measures of one-way ANOVA on the density of cells of AGR treatment revealed a significant difference among the groups in the striatum ($F(3,68) = 6.808$, $P < 0.001$) and in the cortex ($F(3,68) = 4.218$, $P < 0.01$). Post-hoc comparisons indicated that the AGR + ISCH group had more cells in the striatum ($P < 0.05$), but not in the cortex ($P > 0.355$) compared with those of the SAL + ISCH group. There was, however, no significant difference between the URE + ISCH and SAL + ISCH groups in the striatum ($P > 0.104$) and the cortex ($P > 0.566$) as shown in Fig. 8.

Discussion

The present results demonstrated that cerebral ischemia produced severe deficits in performance on the radial arm maze and Morris water maze as well as neuronal cell loss in the hippocampus, striatum and cortex. Pretreatment with AGR and URE attenuated ischemia-induced learning and memory deficits in the Morris water maze and radial arm maze and had a protective effect against ischemia-induced cell damage in the brain.

AGR and the URE are well-known major oriental medicines which have long been included in medical prescriptions for the treatment of stroke and vascular dementia. Although prescriptions including these drugs are clinically known to be effective for stroke-induced cognitive impairments, their therapeutic actions have not been investigated. Recently, several studies have indicated ameliorative effects of these drugs on learning and memory impairments. For example, AGR inhibited scopolamine-induced amnesia in rats (Hsieh et al., 2000). Some oriental prescriptions including AGR have been shown to improve avoidance performance on learning and memory tasks (Nishiyama et al., 1994b; Zhang et al., 1994) and to ameliorate impairment of learning and memory produced by thymectomy or lesioning of the amygdala (Nishiyama et al., 1994a; Zhang et al., 1994). The present findings demonstrated that cerebral ischemia produced severe deficits in performance on the radial arm maze and Morris water maze, and AGR pretreatment produced improved performance of learning and memory in the two tasks. These learning tasks examine different aspects of the learning and memory process. Morris water maze task is thought to test permanent spatial learning capability and reference memory (Morris et al., 1982). The eight-arm radial maze task is another test for spatial learning ability and is thought to indicate not only long-term reference memory but also short-term working memory (Liang et al., 1997). The present study demonstrated that forebrain ischemia impaired behavioral performances in the Morris water maze and radial arm maze, consistent with previous studies (Okada et al., 1995; Olsen et al., 1994; Sopala and Danysz, 2001). Our results suggest that AGR and URE improved both aspects of spatial learning capability and short-term working memory and prevented the cell loss in the hippocampal CA1 region produced by ischemia.

It should be noted that regional selectivity of neuronal death occurs following MCA occlusion. The middle cerebral artery-perfused brain areas such as the parietal cortex, hippocampus, and striatum are mainly affected after cerebral ischemia (Ota et al., 1997; Rice et al., 1981). Several research workers reported that MCAO affected mainly the striatum and cortex (Nagasawa and Kogure, 1990) in contrast to the present results. There are several procedural differences which produce different patterns of cell death among the studies. One of the most critical factors is the time of occlusion; the extent of brain damage produced by MCAO is known to be dependent on the degree of the ischemic insult and its duration. It was suggested that the use of different periods of occlusion in the MCAO model can alter the specific areas affected. For example, it was shown that damage produced by mild injury (30 min of MCAO) was confined to the striatum or cortex, as was reported by others (Nagasawa and Kogure, 1990), whereas more than 2 hr exposure to MCAO led damage of the striatum, cortex and more remote areas including the hippocampus. The exact reasons underlying these differences in susceptibility between regions are not well known (Butler et al., 2002). In our study, more severe injury resulting from 2 hr of MCAO produced cell death in the hippocampus as well as the striatum and cortex. The cell death is of two types; early necrotic cell death in the ischemic core and delayed death of susceptible cells in other neighboring regions. Of the neighboring regions undergoing delayed cell death to ischemic insults, the most vulnerable are found in the hippocampus, which plays a major role in learning and memory (Butler et al., 2002). It was suggested that these types of cell death occur over an extended time, and these cells can be protected by pharmacological treatments, AGR and URE. It is interesting to note that in the present study AGR had more protective effects on hippocamapal cells than on striatal cells. In addition, although treatment with URE tended to have a protective effect on ischemia-induced cell damage in the striatum and cortex, URE exerted no direct effects on striatal and cortical cells.

Supporting these suggestions, recent studies have shown that MCAO produced cortical and striatal infarct as well as damage to the hippocampus, leading to a delayed form of hippocampal cell death as determined by Fluoro-Jade, which has been used as a marker of neuronal degeneration in a number of brain injury models (Butler et al., 2002). Several studies have also shown that MCAO produced increase in extracellular glutamate release, which plays a key role in ischemia-reperfusion neuronal damage through promotion of calcium entry into neurons (Rothman and Olney, 1986). Neuroexcitotoxicity release of glutamate was only observed after prolonged periods (2–4 h) of ischemia (Matsumoto et al., 1996). Histologically, hippocampal CA1 pyramidal cells, which are believed to be glutamatergic or aspartatergic, demonstrated a marked neuronal necrosis on the 5th days after transient ischemia (Matsumoto et al., 1991). The glutamate release in the hippocampus represents the pathophysiological process in the ischemia-reperfusion phase. Future studies are needed to clarify the role of glutamate in the hippocampus after these drug administrations.

There is evidence showing that ischemia-induced cell death in the hippocampal CA1 area is closely correlated with behavioral deficits in learning and memory (Olsen et al., 1994), indicating that the hippocampal CA1 area plays an important role in memory processes involved in the Morris water maze and radial arm maze tasks. In contrast to the current results, MCAO produced long-term spatial cognitive disturbances without any evidence of hippocampal damage (Okada et al., 1995). It is possible that damage to the cerebral cortex and the striatum may be involved in cognitive deficits produced by MCAO. Several lesion studies have shown that the striatum and the cerebral cortex are known to be involved in the process of learning and memory. Cerebral ischemia, which produced selective neuronal damage in the CA1 sector of the hippocampus and in the dorsolateral striatum, has been demonstrated to induce a deficit in spatial learning as measured by an increase in escape latency and swim distance in the Morris water maze (Okada et al., 1995; Olsen et al., 1994). Although either a hippocampal or striatal lesion can affect spatial learning in the water maze (Block et al., 1993; Morris et al., 1982), the reduction of neuronal damage in the hippocampus rather than the striatum is known to be the basis for the reduction of memory performance in the Morris water maze (Block et al., 1997). The increase in swim distance following a complete striatal lesion is accompanied by a decrease in swim speed (Block et al., 1993). It was known that there was no correlation between reduction in striatal damage and in swim distance, suggesting that the striatum is not critically involved in spatial memory. However, analysis of data concerning hippocampal damage and swim distance demonstrates a clear correlation between reduction in hippocampal damage and in swim distance (Block et al., 1997). It is interesting to note that swim speed was not altered by ischemia and was not affected by drug treatment (Block et al., 1997), consistent with our results showing that AGR and URE protected ischemia-induced cell damage mainly in the hippocampus, rather than the cortex or the striatum. Since the hippocampus plays a more critical role in spatial memory than does the striatum, it is more relevant to suggest that the reduction in hippocampal damage due to treatment with these drugs appears to be the basis for the functional improvement in terms of learning and memory.

In addition to behavioral effects, AGR and its major component asarone, (4-[(1E)prop-1-enyl]-1,2,5-trimethoxypropylenobenzene), have a neuroprotective effect against excitotoxic cell death (Cho et al., 2000, 2002). The extract of AGR inhibited glutamate or NMDA-induced excitotoxicity and competed with the glycine binding site of the NMDA receptor, suggesting that the neuroprotective action of AGR or asarone may be through the blockade of the NMDA receptor binding sites (Cho et al., 2000). The NMDA receptor antagonists are proposed to be useful for the treatment or prevention of a variety of neurodegenerative disorders including stroke (Choi, 1992; Muir and Lees, 1995). Therefore, the NMDA receptor antagonist-like actions of the AGR extracts may explain the ameliorative effects on cell damage and on learning and memory deficits produced by ischemia in the present study.

Several studies have also shown that URE has a neuroprotective effect on glutamate-induced cell death (Itoh et al., 1999; Shimada et al., 1999) and Ca2+-blocking inhibitory activity (Shi et al., 2003). Similar to the effect of AGR, application of URE prevented glutamate-induced cell death of cerebellar granule cells (Itoh et al., 1999). Consistent with our results, a recent study showed that a methanol extract of URE protected hippocampal CA1 cells against transient forebrain ischemia produced by 4-vessel occlusion procedure models in rats (Suk et al., 2002). In a pilot dose-response experiment, URE (50, 100, 250 or 1000 mg/kg, i.p.) protected against global 4-vessel occlusion- or MCAO-induced cell death in the hippocampus, and 100 mg of URE produced a maximal neuroprotective effect. The dosage (100 mg/kg) used in this study was a relatively standard dose that other workers had reported in rodent experiments (Sugimoto et al., 2000; Suk et al., 2002). URE contains a variety of the oxyindole and indole alkaloids including rhynchophylline, isorhynchophylline, corynoxeine, isocorynoxeine, geissoschzine methyl ether, hirsuteine and hirsutine, and all of which have been demonstrated to possess vasodilator activity (Yuzurihara et al., 2002). These alkaloids, with isorhynchophylline being the most potent, also have neurorpotective effects on glutamate-induced neuronal death in cultured cerebellar granule cells in rats (Shimada et al., 1999), suggesting that these major components of URE may have been responsible for the protection against ischemia-induced cell damage and behavioral deficits shown in the present study. Future studies are needed to understand the neuroprotective and behavioral effects of these components in the MCAO model.

It has been shown that Choto-san, a main crude drug of URE, prevented impairment of a passive avoidance learning response produced by ischemia or scopolamine injections in mice, possibly through the central serotoninergic system, since this effect was blocked by a serotonin 1A receptor antagonist (Yuzurihara et al., 1999). Several studies have shown that URE has a potent vasodilator effect (Goto et al., 1999, 2000) and inhibited decrease in cerebral blood flow (Sugimoto et al., 2000). This antioligemia effect was eliminated by treatment with an inhibitor of nitric oxide synthase (Sugimoto et al., 2000), suggesting that the activation of nitric oxide synthase by URE contributed to at least part of the improvement in the cerebral circulation. In addition, a recent study showed NMDA receptor antagonist activity of URE in that a URE extract blocked NMDA-evoked currents and reduced NMDA-induced neuronal death (Sun et al., 2003). Similar to neuronal mechanism of AGR, NMDA receptor antagonist-like properties of URE may be responsible for its neuroprotective activity and in reducing the learning and memory deficits produced by ischemia as shown in the present study.

Taken together, the actions of AGR and URE on the serotonin, NO or glutamate systems may contribute to the improvement of neuronal and cognitive deficits produced by ischemia. However, we cannot rule out the involvement of other neurotransmitters such as dopamine or acetylcholine. The present results demonstrated that AGR and URE significantly improved performance on the two learning tasks and prevented cell loss in the hippocampus. They also provide strong evidence that AGR and URE may have potential therapeutic effects on cognitive impairments in clinical patients. However, the mechanism underlying the beneficial effects of these drugs on neural damage and cognitive impairment produced by ischemia should be investigated further.

In summary, the present results demonstrated that cerebral ischemia produced severe deficits in the performance on the radial arm maze and the Morris water maze and neuronal cell loss. Pretreatment with AGR and URE attenuated ischemia-induced learning and memory deficits in the Morris water maze and the radial arm maze and exerted a protective effect against ischemia-induced cell damage in the brain. These results suggest that AGR and URE may be effective in the treatment of vascular dementia.

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